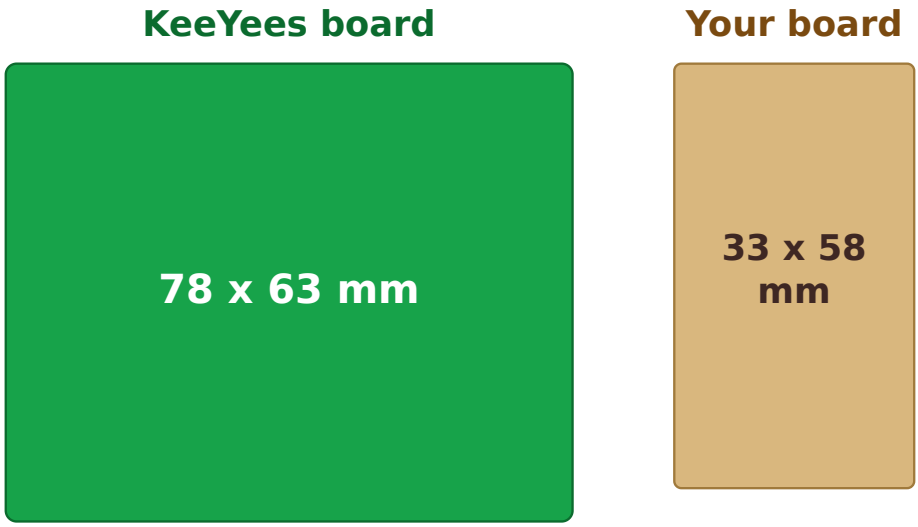
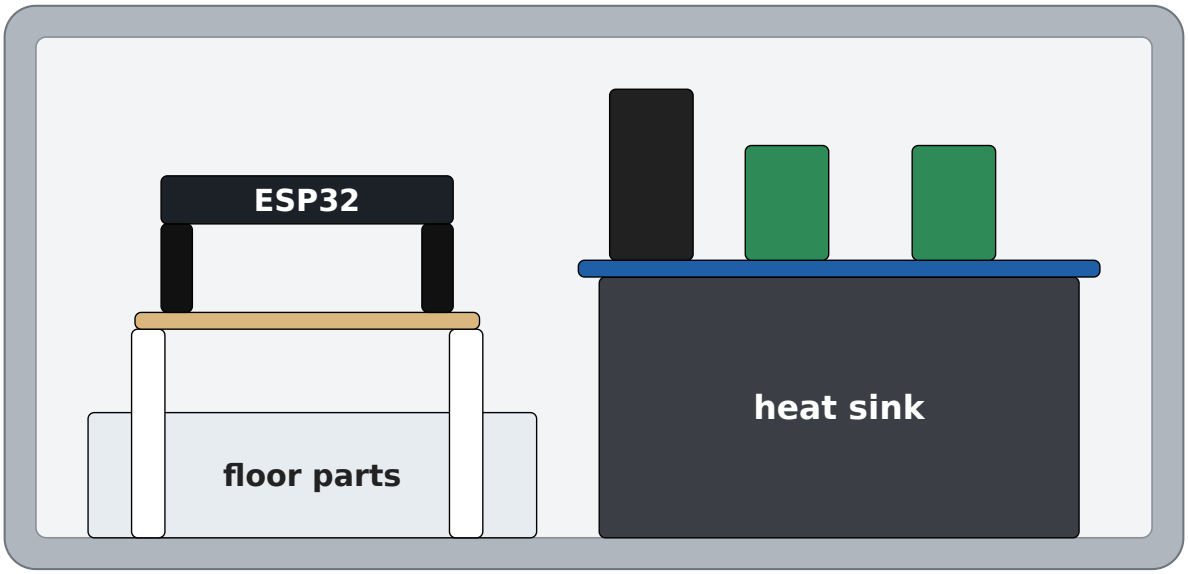
Smaller Awning Box

Your own ESP32 perfboard, 13 x 23 holes, in the 115 x 90 x 55 mm box.



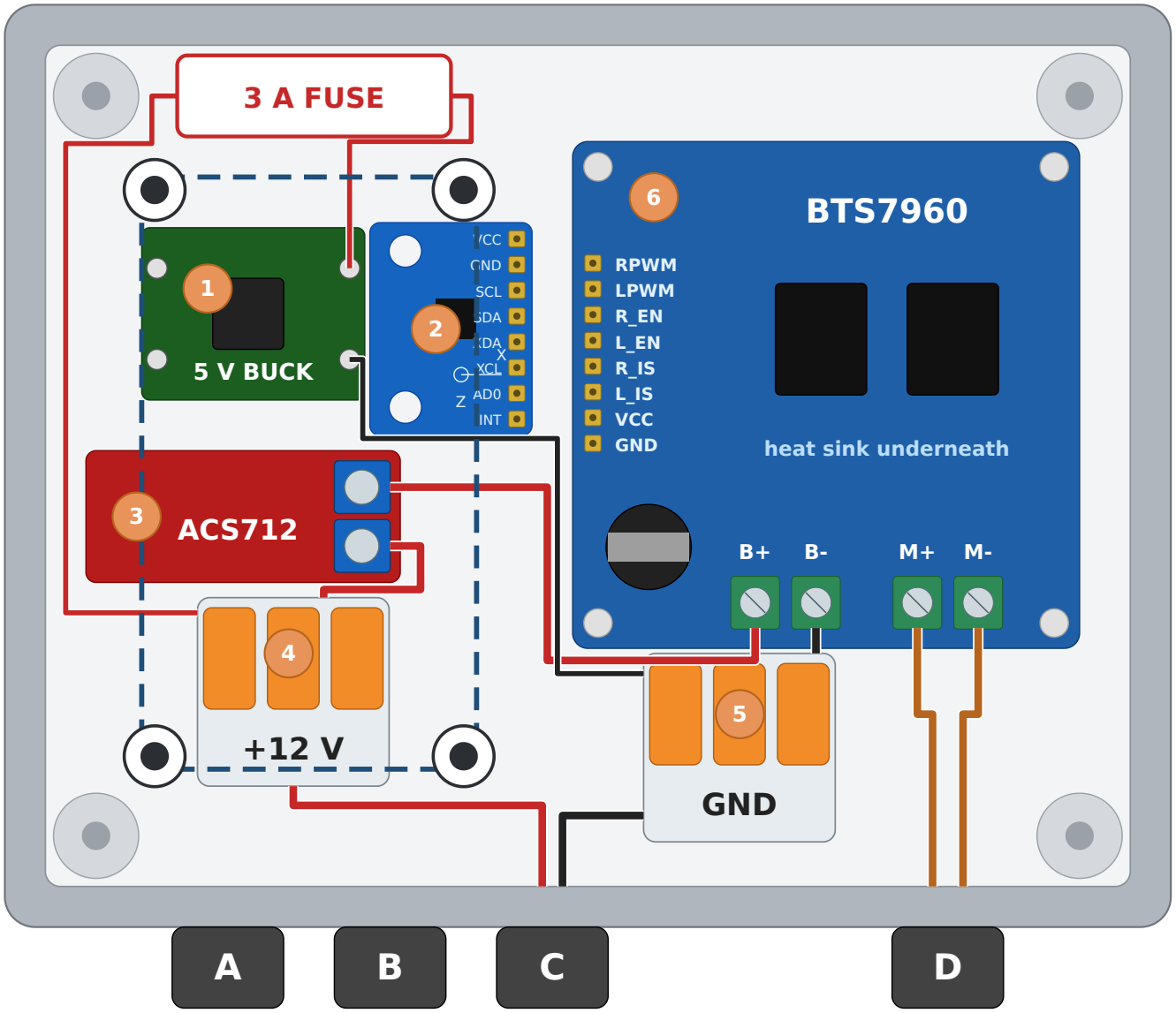
Side view: the small box



Driver: about 43 mm tall, in about 48 mm of space

Check first: the board needs your ESP32's two pin rows to be 10 holes apart. Page 4 shows how to test it in seconds.

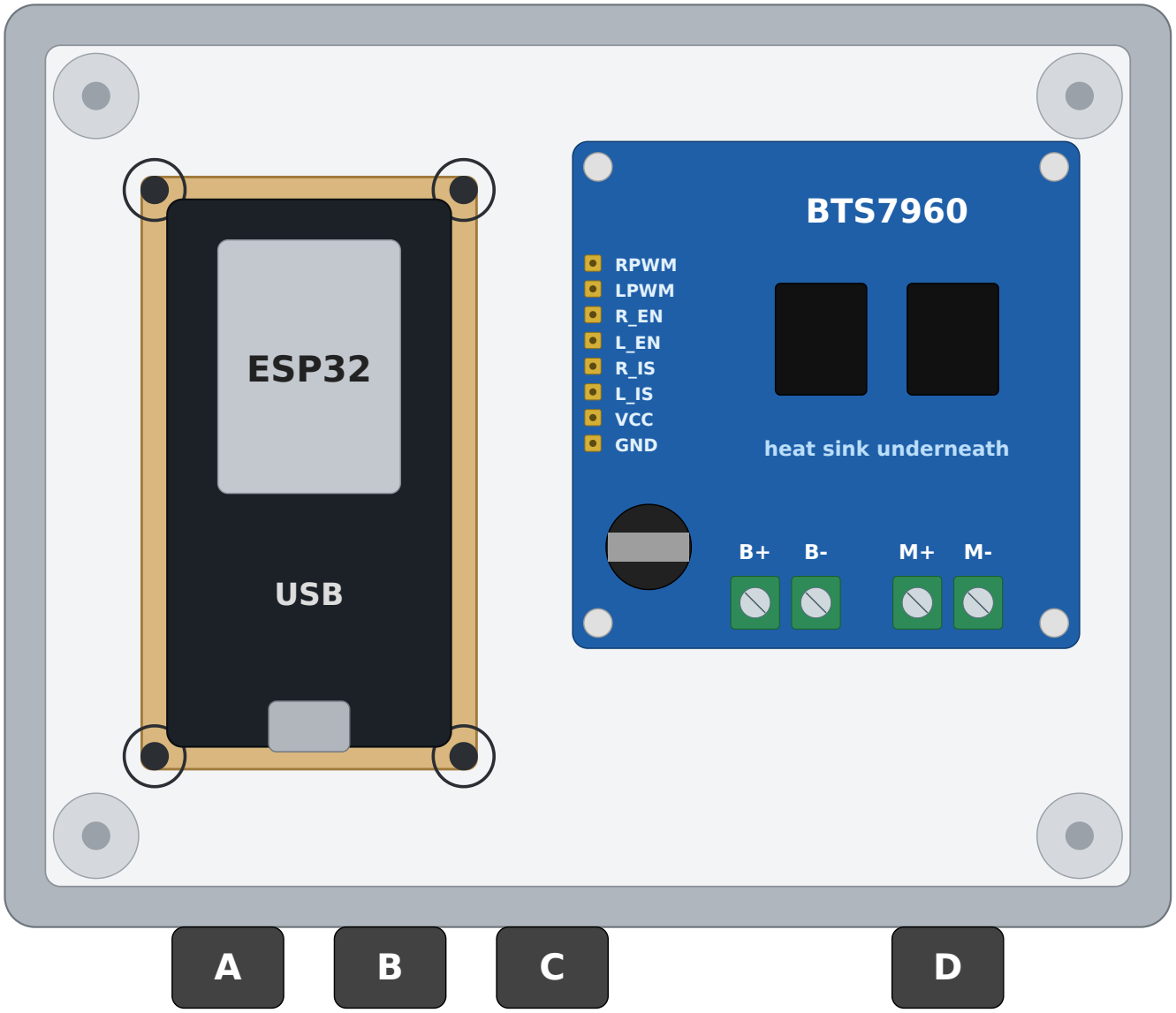
Level 1: the box floor



Floor parts	
1	**5 V mini buck** VHB tape
2	**GY-521 tilt sensor** 2 screws, flat, must not wobble
3	**ACS712 current sensor** VHB tape
4	**Lever nut: +12 V** Zip tie
5	**Lever nut: GND** Zip tie
6	**BTS7960 motor driver** Heat sink on the floor, held firmly

Blue dashed line: where your board sits above these parts.

Level 2: your board, on 20 mm posts



USB toward the glands. Column A, the side with the ESP32 pins you use, faces the left wall.

Wire room: about 9 mm on each side of the board.

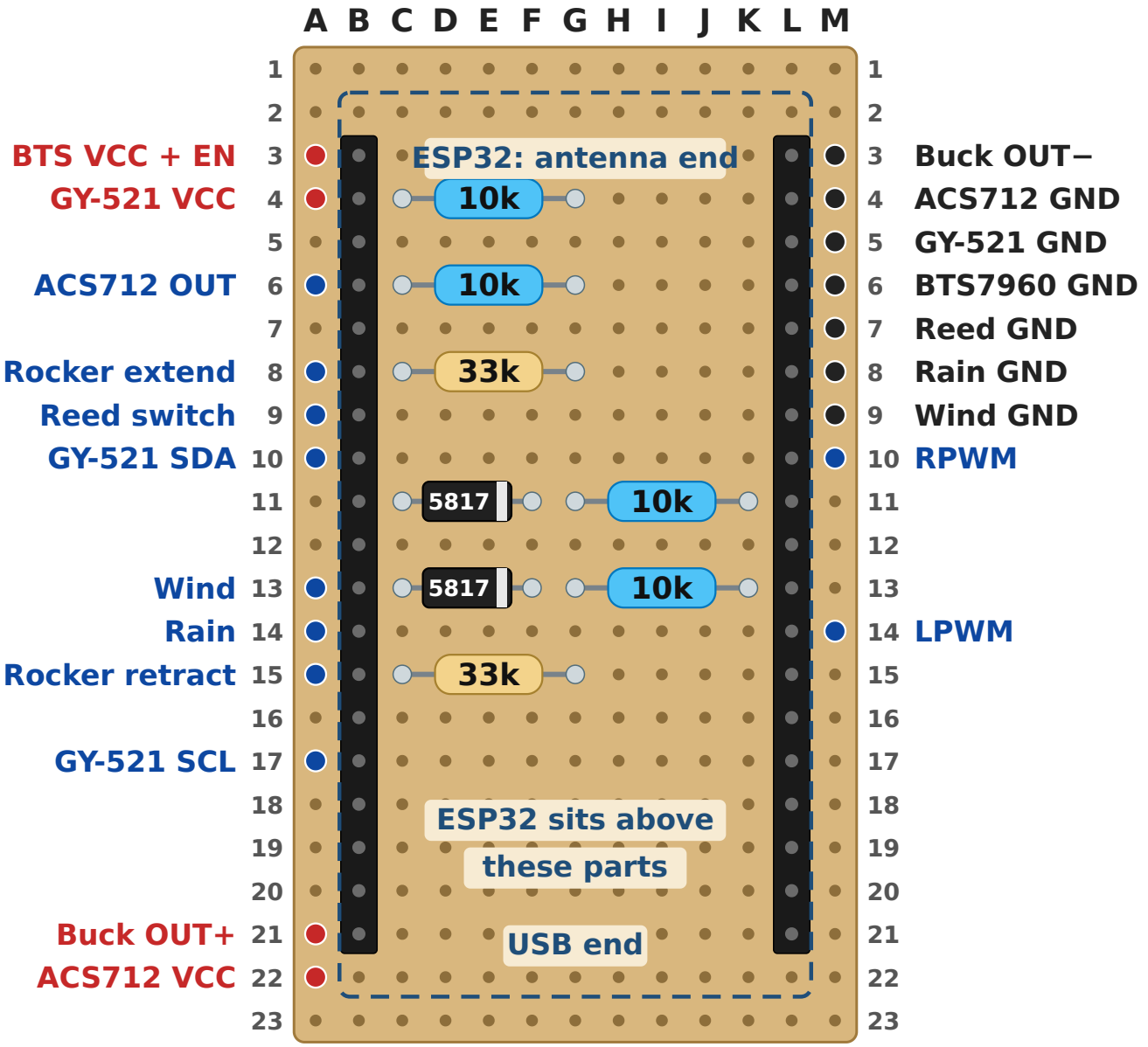
Flashing: update over Wi-Fi. For USB, pull the ESP32 out of its sockets.

Antenna: tape the wire inside the lid, away from the driver.

Test your ESP32 first

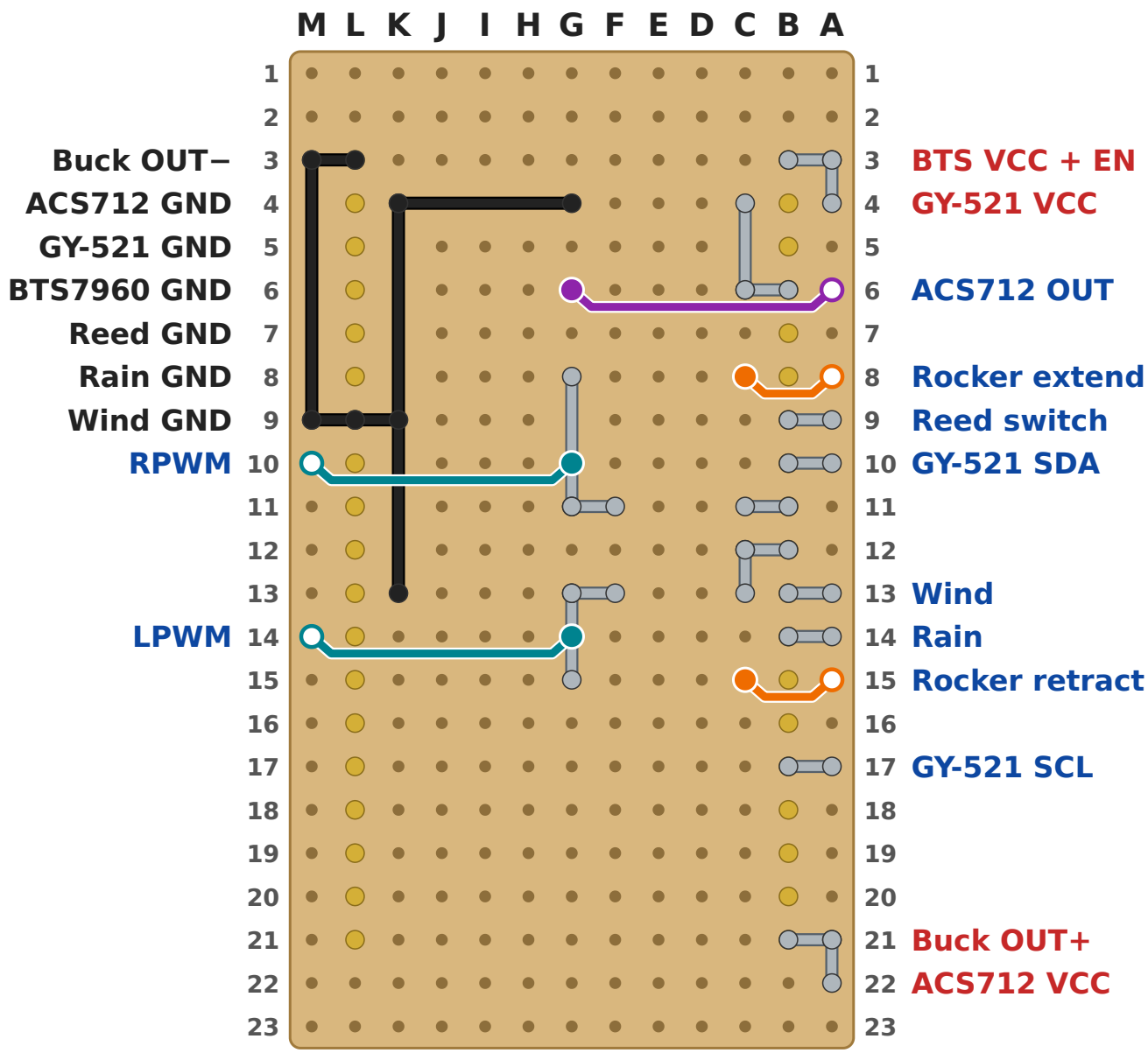
Before you cut anything: set the ESP32's pins into a bare piece of perfboard. They must drop in with 9 empty holes between the two rows, so the rows are 10 holes apart. If they don't, stop and tell me: the layout has to change.

Top of the board: the parts



Underside: the solder side

This view is flipped: column A is on the right. Silver and black lines are bare wire. Colored lines are insulated wire.



Parts on top

Part	Holes
10k	C4 to G4
10k	C6 to G6
33k	C8 to G8
1N5817	C11 to F11, stripe at F11
10k	G11 to K11
1N5817	C13 to F13, stripe at F13
10k	G13 to K13
33k	C15 to G15

Insulated wires underneath

These cross other joints, so keep the insulation on until the last 3 mm. The wire comes over the board's edge, down through its edge hole, and is soldered only at the far end.

Edge hole	Solder at	Wire
A6	**G6**	ACS712 OUT
A8	**C8**	Rocker extend
A15	**C15**	Rocker retract
M10	**G10**	RPWM
M14	**G14**	LPWM

Bare wire underneath

Connection	Join these holes
3V3 pin	A3, A4, B3
GPIO39	B6, C6, C5, C4
GPIO25	B11, C11
GPIO26	B12, C12, C13
GPIO32, 33, 27, 14, 13	Join each pin to the A hole beside it: rows 9, 10, 13, 14, 17
5V pin	A21, A22, B21
Extend signal	F11, G11, then G10, G9, G8
Retract signal	F13, G13, then G14, G15
Ground	L3 to M3. M3 down to M9. M9 to L9 to K9. K4 down to K13. G4 across to K4

Don't bridge neighbors. Column C sits right beside the socket pins in column B, and K sits beside L. Only join the holes listed here.

Outside wires: the other wires come over the edge, down through the hole named on the drawing, and are soldered there. Zip tie outside cables to the box so they can't pull on the board.

Before you plug in the ESP32

With no power on, use your meter's beep setting. 3V3, 5V and every GPIO must **not** beep to ground.

Glands, bottom wall

Gland	Cable
A	4-core cable: reed + wall rocker
B	Optional rain + wind. Leave plugged until used
C	+12 V and ground in
D	Motor

Drill the glands about 35 mm off the floor. Keep each hole clear of the lid seam.

Parts to get

Part	Details
IP65 box	Otdorpatio 115 x 90 x 55 mm with ears. Listed inside: 107 x 83 x 48 mm
Perfboard	2.54 mm holes, cut to 13 x 23 holes
Sockets	2 rows of 19-pin female header
Small parts	Four 10k, two 33k, two 1N5817
5 V mini buck	MP1584 style, up to 22 x 17 mm
M3 nylon posts + screws	20 mm x 4. Drill the 4 corner holes to 3 mm
Lever nuts	WAGO 221-413 x 2
Fuse	3 A with a slim inline holder
Glands	4, about 16 mm across the nut, plus 1 plug

Check before you drill the box

Check	Layout needs
BTS7960, heat sink bottom to tallest part	Listed 43 mm. Must be under 48 mm with wires
Box inside, floor to closed lid	Listed 48 mm
Box inside, length and width	Listed 107 x 83 mm
ESP32 length and width	Drawn 54 x 28 mm
Pin names on your ESP32	Must match the order on the top view. Follow your board's labels

Mini buck warning: many are adjustable and can come set above 5 V. Before connecting anything, power it from 12 V alone and check for about 5 V with your meter. Too much voltage can destroy the ESP32.

Holding the driver down: it stands on its heat sink, with no posts. It must not move on the shaking arm, so fix it firmly and test by hand before closing the lid.

Heat check: this is a tight, sealed box. On the bench, run the motor as hard as a real awning cycle, then feel the driver's heat sink with the lid off.

Before you drill the arm, hold the box in place and fold the awning closed. It must clear the rail, the inner arm and the gas strut.